autoscaling to build high availability for a single virtual machine, as described here:

- Allocate an Elastic IP, and associate this public IP address during the bootstrap of the virtual machine.
- Create or update a DNS entry linking to the current public or private IP address of the virtual machine.
- Use an Elastic Load Balancer (ELB) as a static endpoint that forwards requests to the current virtual machine.

To use the second solution, you need to link a domain with the Route 53 (DNS) service; we've chosen to skip this solution because you need a registered domain to implement it. The ELB solution is covered in chapter 14, so we'll skip it in this chapter as well. We'll focus on the first solution: allocating an Elastic IP and associating this public IP address during the virtual machine's bootstrap.

Execute the following command to create the Jenkins setup based on autoscaling again, using an Elastic IP address as static endpoint:

```
$ aws cloudformation create-stack --stack-name jenkins-multiaz-efs-eip \
➥ --template-url https:/ /s3.amazonaws.com/\
➥ awsinaction-code3/chapter13/multiaz-efs-eip.yaml \
➥ --parameters "ParameterKey=JenkinsAdminPassword,
➥ ParameterValue=$Password" \
➥ --capabilities CAPABILITY_IAM
```

You can find the CloudFormation template for this example on GitHub and on S3. Download a snapshot of the repository at https://github.com/AWSinAction/code3/archive/main.zip. The file we're talking about is located at chapter13/multiaz-efs-eip .yaml. On S3, the same file is located at https://s3.amazonaws.com/awsinaction-code3/chapter13/multiaz-efs-eip.yaml.

The command creates a stack based on the template shown in listing 13.5. The differences from the original template spinning up a Jenkins server with autoscaling follow:

- Allocating an Elastic IP
- Adding the association of an Elastic IP to the script in the user data